Fiscal Year:	FY 2027	**Task Last Updated:**	FY 04/19/2026
PI Name:	Riaz, Amir Ph.D.		
Project Title:	Pool Boiling Heat Transfer Mechanisms in Low Gravity: Numerical Experiments of MABE and NPBX Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ariaz@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-0707
Organization Name:	University of Maryland		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	127 Glenn L Martin Hall		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
Start Date:	12/01/2020	**End Date:**	11/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 12/4/24) NOTE: End date changed to 11/30/2024 per PI (Ed., 11/27/23) NOTE: End date changed to 11/30/2023 per NSSC information (Ed., 11/22/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balaras, Ilias Ph.D. (George Washington University)		
Grant/Contract No.:	80NSSC21K0470		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The key challenge in developing a proper understanding of nucleate boiling in thermal systems for space environments is the accurate characterization of bubble dynamics and associated heat transfer mechanisms over a wide range of wall superheat and liquid subcooling conditions. To address this challenge, we are proposing well-characterized numerical experiments using high fidelity Direct Numerical Simulation for investigating the dynamics of nucleate boiling across a range of gravity conditions. Our numerical experiments will recreate the conditions of the Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) that were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. The focus of NPBX was the investigation of how bubbles grow and merge in microgravity. The main finding was that heat fluxes for steady state nucleate boiling were found to be substantially lower than what was expected on the basis of low gravity data obtained on Earth. The numerical solver that we will use to simulate MABE conditions has been recently validated by comparing the heat flux from our fully 3-D simulations to MABE data for a representative sample of wall superheat and liquid subcooling conditions. Details about our numerical methods and validation case studies have been recently published (Akash Dhruv, Elias Balaras, Amir Riaz, and Jungho Kim. A formulation for high-fidelity simulations of pool boiling in low gravity. International Journal of Multiphase Flow, 120:103099, 2019). These case studies demonstrate that our solver is able to resolve fully the complex interactions of liquid and vapor phases involving phase transition, interphase mass flux, and contact line dynamics. Using the data from the simulations, we have been able to connect the dynamics of bubble nucleation and growth to the corresponding heat flux measurements in MABE data. Such detailed characterizations of the flow field have already started to shape and improve our understanding of the underlying bubble dynamics. In the proposed research, we aim to systematically probe the MABE database now at a deeper level to extract fundamental information on: (i) how is heat flux partitioned into contributions arising from nucleate boiling and natural convection? (ii) what is the role of contact angle in affecting the force of wall adhesion? and (iii) how much heat transfer occurs through microlayers in low gravity settings? These questions will be addressed to qualitatively different regimes of bubble dynamics under various conditions of gravity, wall superheat, and liquid subcooling. The answers to these questions, investigated on the basis of a validated solver using MABE data, would extend the general understanding of pool boiling in low gravity settings. The use of the MABE database is instrumental to this endeavor. It will be used to provide the reference dataset that connects heat flux, and in some cases even visualization of nucleate boiling, to different levels of gravity, wall super heat, and subcooling. This will allow a unique opportunity for a quantitative assessment of potential improvements in the design of thermal systems in space applications. The proposed research is expected to complement NASA's ongoing efforts, in addition to advancing the science of heat flux partitioning, contact line dynamics, and the role of microlayers for nucleate boiling in low gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Electronic devices require effective and efficient means of thermal management to achieve performance targets. The use of latent heat removal via the process of phase change, such as nucleate and pool boiling, is an extremely effective method of thermal management. This method relies primarily on the dominance of buoyancy over capillary forces to maintain high heat flux at the heated surface. The proposed research will study the influence of major parameters on the behavior of boiling heat transfer. The resulting data and analysis will help improve the thermal management of electronic devices, enabling the achievement of higher performance goals.
Task Progress:	The main finding, related to the effects of biphilic surfaces on pool boiling, is that the heat transfer in partial gravity exceeds that in Earth gravity by a wide margin, pointing to a new paradigm for thermal management in partial gravity. Moreover, a peak in the heat transfer rate occurs for specific biphilic patterns, indicating the optimization potential of biphilic surfaces in both Earth and partial gravity. Our investigation has further revealed the fundamental thermal-fluid dynamic mechanisms associated with enhanced thermal performance of pool boiling on biphilic surfaces, pointing to a clear path for achieving optimal performance.
Bibliography Type:	Description: (Last Updated: 04/23/2026)